

App to guide surgery?!

The Fraunhofer iPad app uses augmented reality in order to guide doctors during liver surgery. <http://dgit.in/LiverApp>



Project Spark

Showcased at Gamescom 2013, Project Spark is a game maker in the form of a game itself for the Xbox 360 and the Xbox One. <http://dgit.in/ProjSpark>

MFDs Test

HOW WE TESTED

The MFDs we received were put through an exhaustive test process that consumed many hundreds of pages as we tried to comprehensively account for every possible use-case of an MFD in a SoHo environment. Here's a run-down of the tests we ran:

Initial preparation:

- After installation, the printer is made to run through a complete round-up of diagnostics and checks, including calibration and cartridge alignment, etc. where applicable.
- Fresh toners and cartridges were used for all tests.
- All printer settings were reset to defaults where applicable.
- Laser printers were made to print 10 pages of text so as to warm up before the actual testing began.
- For quality tests, we used a loupe to examine the print quality.

Image Quality:

- We used three different images for each test. One full-colour image, one monochrome image (black) and one calibration image to check printer resolution.
- Certain areas were identified on each image and printers were graded on their ability to render those regions correctly.

Text Quality:

- We used four test files to inspect text quality. The first was a standard text

document in MS Word in 12pt Calibri font. The second was an MS Word document with fonts ranging from 2pts to 12pts in Calibri. The third was a PDF with fonts ranging from 3pts to 20pts and yellow text on a black background with diminishing font size. The final document contained a mixture of plain text at various fonts, coloured text on a coloured background, text embedded in images and text in different orientations and shades.

- Various points were identified on all these documents and rendering ability was checked. Text prints were primarily rated on ink-bleed, sharpness, white spots and merged characters.

Copy Tests:

- For this test we used a print of the first and the fourth text file mentioned above.
- Output was taken at default printer settings and the copy was rated as per the text quality test mentioned above.

OCR Test:

- In this test we used two files. The first was a print of first text file mentioned above and a page from a text-book on computer programming.
- We used ABBYY Fine Reader to analyze the scan and determine the text. Final score was given based on the number of errors and missed text blocks.

Scan test:

- For this test we used a cover of Digit at 300dpi and rated the final image based on each MFDs ability to render certain pre-defined zones accurately.

Speed tests:

- Print speed tests were measured as the time taken to print a pure text file, image file and a mixed document at quality and draft settings on the printer. Scores were individually noted and assigned appropriate weights.
- Warm-up time was measured as the time to print 5 pages from the sleep and power-off states of an MFD where applicable.

Features:

- All salient features of the MFDs were listed out and rated with appropriate weightages.

Cost of ownership:

- Cost of ownership (C) is the cost of running the printer for 3 years and is the total investment required per page that you expect to print. So say you need to print 1000 pages and a cartridge gives you 500 prints, then $C = (\text{Cost of printer} + \text{cost of 2 cartridges})/1000$
- For colour printers, we assumed cost of ownership based on 60% black and white prints and 40% colour prints on 3000 prints.
- For MFDs we used two load scenarios as can be seen from the table overleaf.

large the same. You will see from the table that the performance of these printers does vary considerably, given our scoring, but please bear in mind that these scores are given with the harshest of mentalities while the test prints were examined under a loupe, but the untrained eye will not notice any significant variation in quality.

Ricoh

Ricoh sent us three printers, the SP100U, SP200SU and the SP203SFN. In order, they represent the three basic types of MFDs that you can purchase, namely, a basic laser MFD for low volume printing,

a more advanced MFD that can handle more paper and the most advanced which includes an Automatic Document Feeder and even FAX support. In terms of design these printers are very staid and simple and are exactly like almost every other MFD that we've come across. One unusual design decision is the paper input tray, which is longer than the width of the printer and so, juts out the back. What impressed us most with these printers however was the consistency of the performance; you can pick up either of these printers and rest assured that the print

quality will be consistently above average.

We did notice that the printer struggled when it came to printing large image files at the maximum claimed quality (1200x600dpi) and we



Ricoh SP200S

can only guess at the cause. That said, you'll never really need to print at 1200x600 as prints at 600x600 look the same anyway. All three devices showcased impressive scanning capabilities and while the resolution might be low (compared to scanners on photo printers), the quality and colour tone were excellent.

Where these printers really take a beating is in the cost of ownership with the SP203SFN coming in at an astronomical ₹4 per word and the others at almost ₹3. Another area where these device suffer is in the copy